

Pixxel should prioritize PMFBY crop-insurance analytics via a SaaS-led hybrid model to capture a \$102M India TAM by FY2028

CLIENT SITUATION

- Pixxel has \$95M institutional funding, 3 Firefly hyperspectral satellites operational, and a ₹1,200+ crore / 12-satellite national EO Public-Private Partnership with IN-SPACe (consortium: Dhruva Space, PierSight, SatSure).
- Aurora analytics platform is live; company fragmented across agriculture, mining, oil & gas, maritime — no dominant monetization strategy.
- Window narrowing: Planet Labs expanding, ISRO/Bhuvan provides free baseline EO, domestic startups scaling — Pixxel must transition from hardware-centric startup → profitable data/analytics enterprise.

THE PROBLEM

"Pixxel possesses world-class hyperspectral infrastructure but lacks a concentrated, scalable commercial strategy to convert India's agriculture opportunity into sustainable FY2026–28 profitability."

HYPOTHESIS TESTED

"PMFBY satellite index underwriting is the faster volume path; enterprise precision-ag SaaS is the margin path — best delivered through a SaaS-led hybrid bundling Aurora with standardized imagery."

✓ **HYPOTHESIS CONFIRMED** (market size + regulation) | **QUALIFIED** on competition (SatSure channel strategy required)

THREE RECOMMENDATIONS

01 MARKET

Size and pursue \$102M TAM (base) for satellite-based precision ag analytics in India. PMFBY sub-segment \$72M (71%), enterprise \$30M (29%).
9.18M ha @ \$11.10/ha

02 VERTICAL

Primary: PMFBY / crop insurance index underwriting. Expansion: enterprise precision ag (horticulture FPOs, sugar, export grapes) from FY2027.
8 states PMFBY · 25 enterprise accounts by FY2028

03 MONETIZATION

SaaS-led hybrid: 55% Aurora SaaS / 35% DaaS-ARD / 10% services by FY2028.

~58% blended gross margin

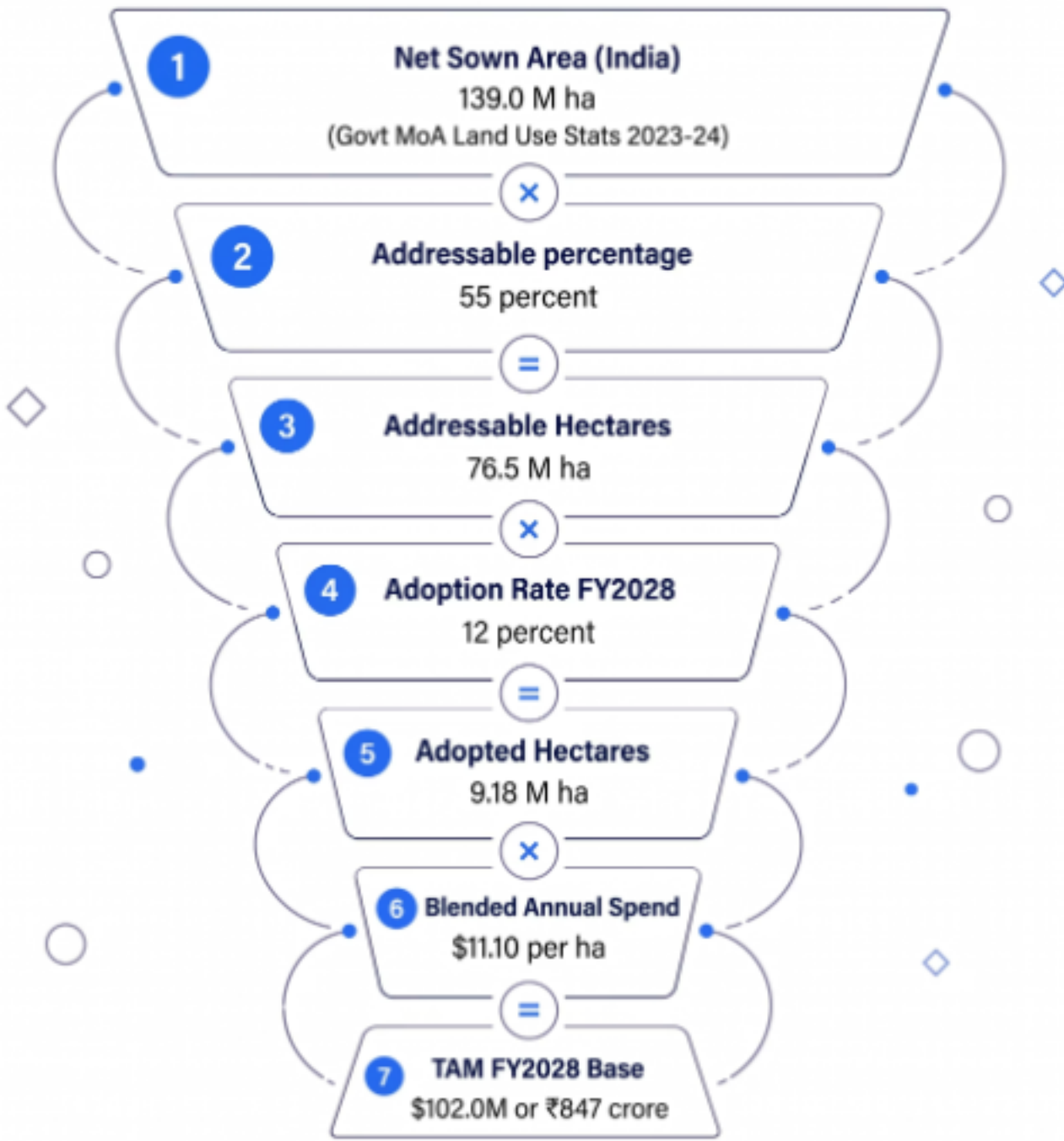
PIXXEL INDIA AG REVENUE TRAJECTORY

FISCAL YEAR	ADDRESSABLE TAM	PIXXEL SHARE	PIXXEL REVENUE	REVENUE MIX
FY2026	\$68M	4%	\$2.7M (₹22 cr)	45% DAAS45% SAAS10% SVC
FY2027	\$84M	8%	\$6.7M (₹56 cr)	40% DAAS50% SAAS10% SVC
FY2028	\$102M	12%	\$12.2M (₹101 cr)	35% DAAS55% SAAS10% SVC

Source: Pixxel commercial analysis · FY2026–28 planning

India's satellite-based precision ag analytics market reaches \$102M by FY2028, driven by PMFBY mandates and 12% adoption of 76.5M addressable hectares

EXHIBIT 2A – BOTTOM-UP TAM WATERFALL



Addressable 55% = irrigated + notified commercial crops; excludes subsistence rainfed smallholders. FX rate: ₹83/USD (planning rate)

EXHIBIT 2B – THREE SCENARIOS

SCENARIO	ADOPTION	\$/HA/YR	ADOPTED HA	TAM
Conservative	6%	\$6.00	4.59 M	\$27.5M / ₹228 cr
BASE ← RECOMMENDED	12%	\$11.10	9.18 M	\$102.0M / ₹847 cr
Optimistic	22%	\$18.00	16.83 M	\$303.0M / ₹2,515 cr

EXHIBIT 2C – TOP-DOWN TRIANGULATION

IMARC Subset Method

\$334.2M × 30% satellite × CAGR adjustment
Result: \$129M (+26% vs base)

✓ SANITY CHECK PASS

Ag GVA Share Method

\$648B ag GVA × 0.016%
Result: \$104M (+2% vs base)

✓ SANITY CHECK PASS

Segment Build Method

\$72M PMFBY + \$30M enterprise
Result: \$102M (= base)

✓ EXACT MATCH

Sanity Check: PASS – Bottom-up base is within 26% of all top-down methods (well under 3× divergence threshold)

Market is large enough to anchor a national strategy — full TAM ≠Pixxel revenue

71% of TAM is PMFBY/insurance (volume game); 29% is enterprise (margin game)

Hyperspectral premium applies to <15% of hectares — multispectral suffices for bulk index monitoring

PMFBY index underwriting outcores enterprise precision ag (3.90 vs 3.55 weighted) on market size and regulatory tailwinds; enterprise expansion follows FY2027

EXHIBIT 3A – WEIGHTED SEGMENT COMPARISON MATRIX

CRITERION	WEIGHT	PRECISION AG	PMFBY INSURANCE
Market Size	25%	3/5 · \$30M sub-TAM	5/5 · \$72M sub-TAM
Willingness to Pay	25%	4/5 · High per account	3/5 · Budget-driven
Competition	20%	3/5 · Planet, drones, Bhuvan	2/5 · SatSure incumbent
Regulatory Framework	30%	4/5 · Light touch	5/5 · YES-TECH + PPP mandate
WEIGHTED TOTAL	100%	3.55	3.90 ✓ WINNER

EXHIBIT 3B – SEGMENT SCALE COMPARISON

Enterprise Precision Ag - FY2028 sub-TAM: \$30M - Buyer count: ~200 accounts - Avg contract: \$150K ARR - Key competitor: Planet Labs - Regulatory catalyst: Digital Ag Mission - Verified scale: Sahyadri FPO ₹1,549 cr turnover	PMFBY Insurance - FY2028 sub-TAM: \$72M - Buyer count: ~600 district-season contracts - Avg contract: \$120K/season - Key competitor: SatSure (consortium partner) - Regulatory catalyst: YES-TECH 30% weight - Verified scale: 4.19 crore farmers enrolled
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Win condition: 2 state PMFBY contracts signed by end FY2026

EXHIBIT 3C – FY2026–28 EXPANSION SEQUENCE

FY2026 – PROVE - Primary motion: PMFBY pilots + 3 enterprise proofs - Targets: 2 states, 3 crops; Sahyadri-type FPO pilots - Revenue mix badge: 70% insurance / 30% enterprise - KPI: \$2.7M revenue target
FY2027 – SCALE - Primary motion: PMFBY expansion + enterprise beachhead - Targets: 6 states; 10 enterprise accounts - Revenue mix badge: 60% insurance / 40% enterprise - KPI: \$6.7M revenue target
FY2028 – MARGINS - Primary motion: Aurora SaaS upsell at scale - Targets: 8 states PMFBY; 25 enterprise; national EO live - Revenue mix badge: 50% / 50% - KPI: \$12.2M revenue target

SaaS-led hybrid (55% Aurora / 35% DaaS / 10% services) maximizes blended margin — pure DaaS commoditizes; pure SaaS misses PMFBY scale

~58%

Blended Gross Margin

FY2028 target for the SaaS-led hybrid mix

The hybrid model balances Aurora SaaS, DaaS volume, and services to improve margin while preserving PMFBY scale.

EXHIBIT 4A — UNIT ECONOMICS COMPARISON

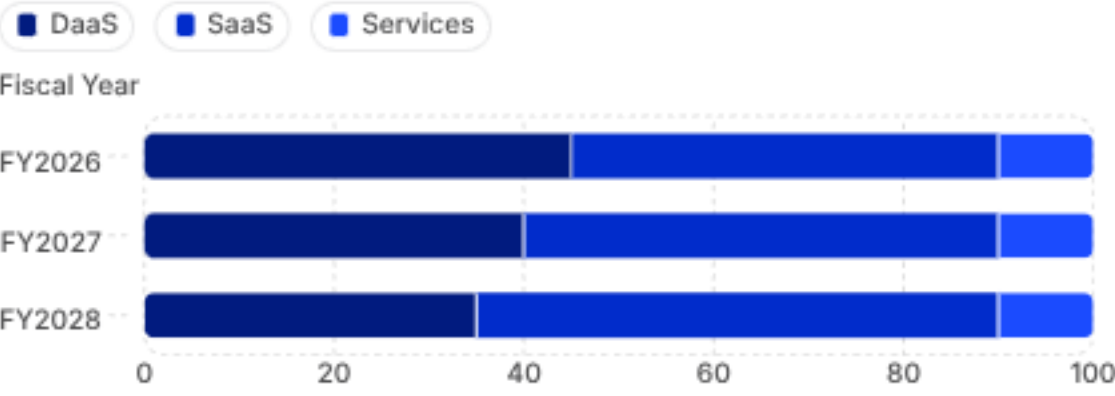
DaaS (ARD Bulk) - Gross Margin: 45% - Typical deal: \$1.2M / state-season - CAC: \$80K (tender process) - LTV:CAC: 30:1 - Payback: 2.0 years - Scalability: Very High (hectares) - Concentration risk: HIGH — top 3 accounts = 60% Small example: "2M ha × \$0.60/ha = \$1.2M revenue · 45% GM = \$540K gross profit"	SaaS (Aurora) - Gross Margin: 70% ← highlighted - Typical deal: \$300K / enterprise ARR - CAC: \$45K (ABM) - LTV:CAC: 17:1 - Payback: 1.1 years - Scalability: Moderate (accounts) - Concentration risk: LOWER (20+ accounts) Small example: "8 Aurora Enterprise users + imagery + analytics = \$300K ARR · 70% GM = \$210K gross profit"	RECOMMENDED Hybrid FY2028 - Blended Gross Margin: ~58% - Revenue: \$12.2M - Mix: 55% SaaS / 35% DaaS / 10% Svc - Risk: Mitigated by diversified mix Blended GM Target
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EXHIBIT 4B — PUBLISHED PRICING ANCHORS (PIXCEL)

PRODUCT	PRICE
Hyperspectral archive imagery	\$6 /km²
Hyperspectral tasking imagery	\$8 /km²
Aurora Professional	\$150 /user/month
Aurora Enterprise	\$180 /user/month (annual)

Bulk B2G contracts discount per-pass imagery 80–90% but cover more hectares and revisits.

EXHIBIT 4C — REVENUE MIX TRAJECTORY



FY2026: 45% DaaS | 45% SaaS | 10% Svc — total \$2.7M — GM 54% · FY2027: 40% | 50% | 10% — total \$6.7M — GM 56% · FY2028: 35% | 55% (largest section) | 10% — total \$12.2M

COMPETITIVE PRICING CONTEXT

Planet Labs PlanetScope from \$2,700/yr per 50km² → "Commodity monitoring pressure on pure DaaS"	ISRO Bhuvan Free/low-cost gov EO baseline → "Pixxel must sell analytics value, not raw pixels"	SatSure YES-TECH insurance analytics incumbent → "Partner, don't duplicate"
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CONDITIONS TO REVISIT MODEL

Trigger	Action
PMFBY tender margin <30%	Mandate SaaS attach in all bids
Aurora enterprise churn >25%	Shift to DaaS volume
National constellation delayed >12 months	Enterprise SaaS-first
SatSure partnership fails	Build/acquire YES-TECH capability

❏ So-what: Lead with Aurora SaaS (70% GM) · Use DaaS for PMFBY volume · Target 58% blended GM by FY2028

Execute 3-phase roadmap — Prove (FY2026), Scale (FY2027), Margins (FY2028) — targeting \$12.2M India ag revenue at 12% TAM share; gate: 2 PMFBY state wins by end FY2026

EXHIBIT 5A — PHASED GTM ROADMAP

FY2026

PROVE

FY2026 PROVE

- PMFBY Pilots: "Bid 2 state YES-TECH pilots (paddy/wheat + 1 horticulture)" · KPI: 2 signed contracts
- Enterprise: "3 Aurora pilots: 1 FPO (Sahyadri), 1 sugar co-op, 1 input company" · KPI: 3 pilots → 1 paid conversion
- Product: "Ship insurance index dashboard; CCE optimization API" · KPI: YES-TECH 30% weight compliant
- Partnership: "Formalize SatSure data-supply SLA in consortium" · KPI: Signed SLA

FY2027

SCALE

FY2027 SCALE

- PMFBY: "Expand to 6 states; integrate national EO ARD feed" · KPI: 6 active contracts
- Enterprise: "10 paying accounts; hyperspectral premium for horticulture" · KPI: \$1.5M enterprise ARR
- Operations: "Scale ground segment for 12-sat constellation prep" · KPI: 99% data delivery SLA

FY2028

MARGINS

FY2028 MARGINS

- PMFBY: "8 states; automated claims validation module" · KPI: \$8.5M insurance revenue
- Enterprise: "25 accounts; Aurora self-serve for mid-market FPOs" · KPI: \$5.5M enterprise ARR
- Mix: "55% SaaS / 35% DaaS / 10% services" · KPI: 58% blended GM

EXHIBIT 5B — KPI DASHBOARD (BOARD TRACKING)

KPI	FY2026	FY2027	FY2028
India ag revenue (\$M)	2.7	6.7	12.2
Hectares under monitoring (M)	1.5	5.0	9.0
PMFBY states active	2	6	8
Enterprise paying accounts	1	10	25
Blended gross margin	54%	56%	58%
SaaS revenue %	DAAS 45%SAAS 45% SVC 10%	DAAS 40%SAAS 50% SVC 10%	DAAS 35%SAAS 55% SVC 10%
CAC payback (months)	18	14	12

EXHIBIT 5D — IMMEDIATE NEXT 90 DAYS

01

01 Board approval of PMFBY-first strategy and SaaS-led hybrid model

03

03 Sign SatSure data-layer agreement within consortium framework

05

05 Staff govt sales lead with PMFBY/RFP experience

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✓ Win Condition FY2026: 2 PMFBY state contracts signed · \$2.7M India ag revenue · 1 enterprise paid conversion

EXHIBIT 5C — RISK REGISTER

RISK	LIKELIHOOD	IMPACT	MITIGATION
SatSure channel conflict	MEDIUM	HIGH	Consortium SLA; complementary positioning
Govt procurement delay	HIGH	MEDIUM	Parallel enterprise revenue buffer
Planet price undercut	MEDIUM	MEDIUM	Hyperspectral differentiation; analytics bundle
YES-TECH scope limited to paddy/wheat	MEDIUM	HIGH	Advocate MNCFC expansion with classification data
Constellation delay	LOW	HIGH	Firefly satellites bridge FY2026–27
INR/USD volatility	MEDIUM	LOW	Price contracts in INR for domestic

02

02 Identify 2 target states for YES-TECH pilot (shortlist: Maharashtra, Karnataka, MP)

04

04 Launch 3 enterprise Aurora pilots with defined success metrics

06

06 Build FY2026 financial model in board dashboard format (monthly KPI tracking)